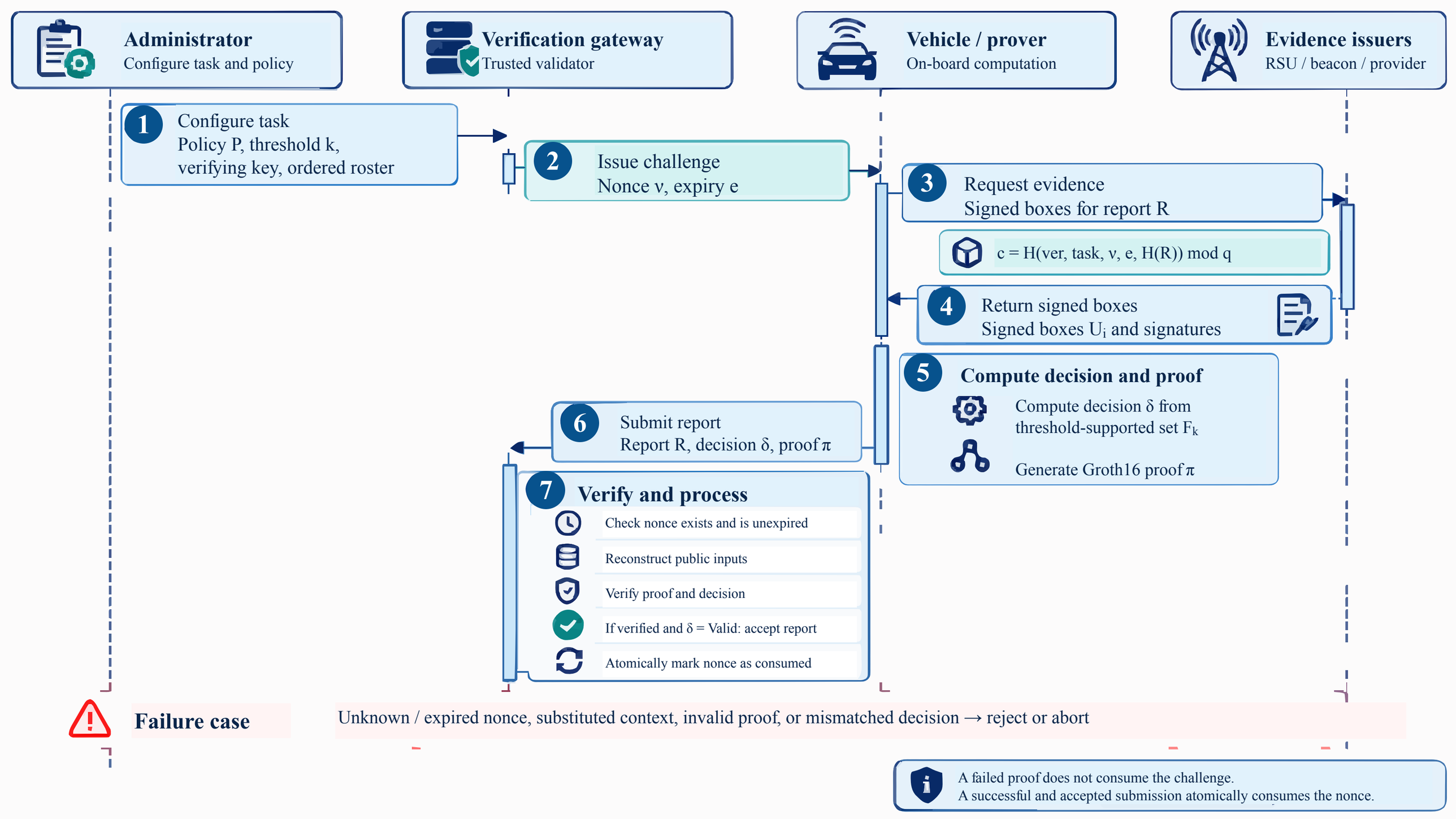

Administrator
Configure task and policy
Verification gateway
Trusted validator
Vehicle / prover
On-board computation
Evidence issuers
RSU / beacon / provider
1
Configure task
Policy P, threshold k,
verifying key, ordered roster
2
Issue challenge
Nonce ν, expiry e
3
Request evidence
Signed boxes for report R
c = H(ver, task, ν, e, H(R)) mod q
4
Return signed boxes
Signed boxes U_i and signatures
5
Compute decision and proof
Compute decision δ from
threshold-supported set F_k
Generate Groth16 proof π
6
Submit report
Report R, decision δ, proof π
7
Verify and process
Check nonce exists and is unexpired
Reconstruct public inputs
Verify proof and decision
If verified and δ = Valid: accept report
Atomically mark nonce as consumed
Failure case
Unknown / expired nonce, substituted context, invalid proof, or mismatched decision → reject or abort
A failed proof does not consume the challenge.
A successful and accepted submission atomically consumes the nonce.